

Automated Ambulatory Blood Pressure Monitoring - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses automated ambulatory blood pressure monitoring.

1. Medical Necessity

Aetna considers automated ambulatory blood pressure monitoring medically necessary according to the selection criteria listed below, which are based, in part, on guidelines developed by the American College of Physicians.

Note: Ambulatory blood pressure monitoring for less than 24 hours or more than 3 days is not considered medically necessary; repeat testing is not generally necessary more frequently than every 6 months.

Member must meet *any* of the following criteria:

1. Office or "white coat" hypertension

Ambulatory blood pressure monitoring is considered a medically necessary preventive service for screening to confirm the diagnosis of hypertension and distinguish "white coat" hypertension or a transient rise of blood pressure that occurs in the office setting. The member must have a minimum of two blood pressure readings in the office setting taken on separate days that are repeatedly elevated (systolic readings of 140 mm Hg or greater and/or diastolic readings of 90 mm Hg or greater in adults, or greater than 90th percentile for age, gender and height in children (see appendix)).

2. Episodic hypertension

Ambulatory blood pressure monitoring is considered medically necessary for members whose symptomatology (paroxysms of excessive sweating, palpitations, apprehension) suggests episodic hypertension secondary to an adrenal tumor (e.g., pheochromocytoma), and office blood pressure measurements are repeatedly normal.

3. Evaluation of hypotensive symptoms

Ambulatory blood pressure monitoring is considered medically necessary for members with

hypotensive symptoms and/or syncopal events that are thought to be related to anti-hypertensive medications.

4. Evaluation of syncope

Ambulatory blood pressure monitoring is considered medically necessary when used in conjunction with a 24-hour Holter monitor (see [CPB 0019 - Holter Monitors](#)) to determine whether symptoms of syncope or near syncope are the direct result of an arrhythmia.

5. Nocturnal angina

Ambulatory blood pressure monitoring is considered medically necessary to investigate blood pressure changes in members with nocturnal angina.

6. Resistant hypertension

Ambulatory blood pressure monitoring is considered medically necessary prior to instituting an invasive investigation (e.g., renin vein assays, angiogram for renal artery stenosis) for secondary causes of hypertension for members with hypertension that is refractory to medications.

7. Autosomal Dominant Polycystic Kidney Disease

Automated ambulatory blood pressure monitoring is considered medically necessary for in children and young adults (30 years of age or younger) with autosomal dominant polycystic kidney disease.

2. Experimental, Investigational, or Unproven

Aetna considers the use of ambulatory blood pressure monitoring experimental, investigational, or unproven in *any* of the following situations because the medical literature does not support its use in these conditions:

1. For blood pressure monitoring of persons with heart failure; or
2. For blood pressure monitoring of pregnant women who do not meet any of the criteria listed above; or
3. For diagnosing malignant (accelerated phase) hypertension. Under accepted guidelines, malignant hypertension requires urgent hospital admission for appropriate investigation and treatment; or
4. For members with an irregular cardiac rhythm (e.g., atrial fibrillation). Blood pressure readings are inconsistent and unreliable when an irregular cardiac rhythm is present due to variances in pulse volume; or
5. For monitoring normal or borderline hypertensive blood pressure readings in the medical setting of members with documented evidence of end-organ damage (e.g., nephropathy, electrocardiographical changes, left ventricular hypertrophy, angina, myocardial infarction, cerebrovascular accident, transient ischemic attack) or cardiovascular risk factors (e.g., diabetes mellitus, smoking, hypercholesterolemia); or
6. For routine monitoring to establish the clinical diagnosis of hypertension or to evaluate the member's blood pressure responses to treatment; or
7. Routine use before initiation of continuous positive airway pressure (CPAP) therapy (i.e., performance of ABPM prior to CPAP in persons who do not have any of the medically necessary indications for ABPM listed in section I).

3. Related Policies

- [CPB 0019 - Holter Monitors](#)
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CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
93784	Ambulatory blood pressure monitoring, utilizing a system such as magnetic tape and/or computer disk, for 24 hours or longer; including recording, scanning analysis, interpretation and report
93786	recording only
93788	scanning analysis with report
93790	physician review with interpretation and report
Other CPT codes related to the CPB:	
36251	Selective catheter placement (first-order), main renal artery and any accessory renal artery(s) for renal angiography, including arterial puncture and catheter placement(s), fluoroscopy, contrast injections(s), image postprocessing, permanent recording of images, and radiological supervision and interpretation, including pressure gradient measurements when performed, and flush aortogram when performed; unilateral
36252	bilateral
36253	Superselective catheter placement (one or more second order or higher renal artery branches), renal artery and any accessory renal artery(s) for renal angiography, including arterial puncture and catheter placement(s), fluoroscopy, contrast injections(s), image postprocessing, permanent recording of images, and radiological supervision and interpretation, including pressure gradient measurements when performed, and flush aortogram when performed; unilateral
36254	bilateral
80416	Renal vein renin stimulation panel (e.g., captopril)
80417	Peripheral vein renin stimulation panel (e.g., captopril)
84244	Renin
ICD-10 codes covered if selection criteria are met:	
D35.00 - D35.02	Benign neoplasm of adrenal gland
I10	Hypertension [malignant only]
I11.9	Hypertensive heart disease without heart failure
I15.0 - I16.9	Secondary hypertension [malignant only]
I20.81 - I20.89	Other forms of angina pectoris [Angina decubitus]

Code	Code Description
I95.0 - I95.9	Hypotension
R03.0	Elevated blood-pressure reading, without diagnosis of hypertension
R55	Syncope and collapse
Q61.2	Polycystic kidney, adult type [autosomal dominant]
ICD-10 codes not covered for indications listed in the CPB:	
E08.00 - E13.9	Diabetes mellitus
E78.010 - E78.019	Pure and familial hypercholesterolemia
E78.2	Mixed hyperlipidemia
F17.200 - F17.299	Nicotine dependence
I09.81	Rheumatic heart failure (congestive)
I21.01 - I25.2	ST elevation (STEMI) and non-ST (NSTEMI) myocardial infarction
I21.A1	Myocardial infarction type 2
I21.A9	Other myocardial infarction type
I50.1 - I50.9	Heart failure
I51.7	Cardiomegaly
I65.01 - I67.9	Occlusion and stenosis of precerebral arteries, not resulting in cerebral infarction
M10.30 - M10.39	Gout due to renal impairment
• N05.0 - N05.9 • N17.1 - N17.2	Nephritis and nephropathy
N17.0 - N17.9	Acute kidney failure
N18.1 - N18.9	Chronic kidney disease (CKD)
N19	Unspecified kidney failure
R03.1	Nonspecific low blood pressure reading
R94.31	Abnormal electrocardiogram [ECG] [EKG]
Z13.6	Encounter for screening for cardiovascular disorders

Code	Code Description
Z34.00 - Z34.93	Encounter for supervision of normal pregnancy

Background

Automated ambulatory blood pressure (BP) monitoring is an outpatient procedure using fully automated devices to measure ambulatory BP at frequent intervals during the day and night in an effort to determine the variability of a patient's BP due to environmental stresses and to aid in definitively establishing a diagnosis of hypertension before committing the patient to life-long antihypertensive therapy.

Continuous ambulatory blood pressure (CABP) monitoring, (also known as noninvasive blood pressure monitoring [NIBP], ambulatory blood pressure monitoring [ABPM] and self-measured blood pressure monitoring [SMBP]) is a noninvasive method of measuring blood pressure (BP) over a specified period of time. Monitoring is generally done for at least 24 hours, but can be up to a maximum of three days. This is a fully automated or semi-automated device consisting of a portable battery-operated monitor worn on the hip (attached to a belt or clothing) and connected to an inflatable blood pressure cuff. The cuff will inflate at pre-set intervals (usually every 15 to 30 minutes during the day and every 30 minutes to an hour at night). The BP is then measured and recorded in the monitor where it can be retrieved for interpretation. In addition to the automatic preset readings, the semi-automated monitor can also be activated to take additional readings such as when the individual is exerting herself/himself or has symptoms of dizziness or heart racing.

CABP monitoring is utilized to evaluate individuals who are suspected of having an elevated BP only in the physician's office (known as white coat hypertension). It has also been proposed for use in monitoring hypotension and the effectiveness of medication. CABP is also purported as a method to predict cardiovascular morbidity or mortality related to left ventricular hypertrophy and may provide useful information regarding the BP in individuals who may have chronic kidney disease, diabetes or preeclampsia. In addition, it could potentially provide the resources to capture a drop greater than 10% in nightly BP, which is called "dipping." Non-dippers are individuals who have less than a 10% decrease in their nightly BP and potentially have a higher cardiovascular mortality rate.

Since treatment is rarely urgent in the absence of severe hypertension, the physician's diagnosis of hypertension should be substantiated first by repeated office readings by well-trained clinicians.

Patients with borderline hypertensive measurements in the office setting should have basic cardiovascular tests done. Those who have evidence of target-organ damage or other cardiovascular risk factors should receive non-pharmacological and/or pharmacological treatments without further investigation. Studies have unequivocally demonstrated that these patients have a significant risk of developing cardiovascular disease and will benefit from antihypertensive therapy. Patients with no evidence of target-organ damage and no risk factors should be classified by a trial of self-measured BP; drug treatment should be considered for patients with consistently elevated readings in this setting.

The U.S. Preventive Services Task Force (2015) recommends screening for high blood pressure in adults aged 18 years or older. The USPSTF recommends obtaining measurements outside of the clinical setting for diagnostic confirmation before starting treatment. The Task Force states that ambulatory blood pressure monitoring and home blood pressure monitoring may be used to confirm a diagnosis of hypertension after initial screening. The USPSTF found convincing evidence that ambulatory blood pressure monitoring is the best method for diagnosing hypertension. Although the criteria for establishing hypertension varied across studies, there was significant discordance between the office diagnosis of hypertension and 12- and 24-hour average blood pressures using ambulatory blood pressure monitoring, with significantly fewer patients requiring treatment based on ambulatory blood pressure monitoring. Elevated ambulatory systolic blood pressure was consistently and significantly associated with increased risk for fatal and nonfatal stroke and cardiovascular events, independent of office blood pressure. For these reasons, the USPSTF recommends ambulatory blood pressure monitoring as the reference standard for confirming the diagnosis of hypertension. The USPSTF states that home blood pressure monitoring using appropriate protocols is an alternative method of confirmation

if ambulatory blood pressure monitoring is not available.

In a systematic review, Goyal and colleagues (2005) stated that "ambulatory blood pressure monitoring has established its use in the definition of white coat hypertension and monitoring of treatment of essential hypertension. Any role for ambulatory blood pressure monitoring in heart failure is not well defined Prospective controlled studies on the impact of treatments on circadian blood pressure profile in congestive heart failure patients are needed".

Sorof and Portman (2000) reviewed their experience using ambulatory BP monitoring in children referred to a hypertension clinic to determine the frequency of pediatric white coat hypertension (WCH), which was defined by 3 different diagnostic criteria:

Clinic BP values were available in 67 otherwise healthy children who underwent ambulatory BP monitoring; 51 had confirmed clinic hypertension by Task Force criteria. Frequency of WCH in these 51 patients with the stated criteria was 53 %, 45 %, and 22 %, respectively. Elevated BP load was found in 52 % (12/23) of patients with normal mean BP. The authors concluded that these findings suggested that many children referred for casual BP elevation have WCH even by strict diagnostic criteria. Ambulatory BP monitoring may help differentiate WCH from persistent hypertension, thereby avoiding unnecessary diagnostic evaluation and identifying children most likely to benefit from early intervention.

Stergiou et al (2004) stated that office and out-of-office BP measurements are being used for the diagnosis of hypertension in children and adolescents. The U.S. National Heart, Lung, and Blood Institute have recently presented a new classification of BP. On the basis of office measurements the 90th, 95th and 99th percentile for gender, age and height are used to classify children and adolescents as normotensive, pre-hypertensive and stage-1 or stage-2 hypertensive. Although auscultation using a standard mercury sphygmomanometer remains the recommended method, accumulating evidence suggests that ambulatory BP monitoring is useful for the detection of WCH and the prediction of target organ damage in children and adolescents. Studies have shown ambulatory BP to be more reproducible than office measurements and normative tables for ambulatory measurements have been developed from cross-sectional studies in children and adolescents. In regard to home measurements in children, there are limited data from small trials showing lower BP levels than daytime ambulatory BP. The authors concluded that ambulatory BP monitoring is already finding a role as a supplementary source of information in children and adolescents, whereas at present home measurements should not be used for decision making in this population.

In a review and meta-analysis, Bliziotis et al (2012) examined the association of home BP measurements with target organ damage. A PubMed and Cochrane Library search (1950 to 2011) revealed 23 studies reporting comparative data of home BP versus ambulatory and/or office measurements in terms of their association with several indices of target organ damage. Correlation coefficients were pooled by random-effects model meta-analysis. A total of 14 studies (n = 2,485) assessing echocardiographic left ventricular mass index (LVMI) showed similar correlations with home (coefficients $r = 0.46/0.28$, systolic/diastolic) as with ambulatory BP ($0.37/0.26$, $p = \text{NS}$ for difference versus home BP), and superior to office measurements ($r = 0.23/0.19$, $p < 0.001/0.009$ for difference versus home BP). Four methodologically heterogeneous studies assessing the glomerular filtration rate (n = 609) could not be pooled or lead to a concrete result. Four studies assessing carotid intima-media thickness (n = 1,222), 3 assessing pulse wave velocity (n = 720) and 2 assessing urinary protein excretion (n = 156) showed no difference in pooled correlation coefficients with home versus office BP measurements. With all the measurement methods, systolic blood pressure (SBP) was more closely associated with target organ damage than diastolic blood pressure (DBP). The authors concluded that these data suggested that home BP is as good as ambulatory monitoring and superior to office measurements in regard to their association with pre-clinical organ damage assessed by echocardiographic LVMI. They stated that more research is needed to evaluate the relationship of home BP with other indices of target organ damage.

Swartz et al (2008) determined the cost-effectiveness of ambulatory BP monitoring in the initial evaluation of stage 1 hypertension. Retrospective chart review of data for children referred to Texas Children's Hospital hypertension clinic between January 2005 and August 2006 was performed. These investigators compared the costs of standard evaluations versus the initial use of ambulatory BP monitoring for children with clinic BP measurements suggesting stage 1 hypertension. Charges for clinic visits, laboratory tests, and imaging were obtained from the Texas Children's Hospital billing department.

A total of 267 children were referred – 139 children did not receive ambulatory BP monitoring; 54 met clinical indications for ambulatory BP monitoring but did not receive it because it was not a covered expense (44 children) or the family refused the study (10 children); 126 children received clinically indicated ambulatory BP monitoring, paid for either through insurance or by the family. Fifty-eight children (46 %) had confirmed white-coat hypertension, 62 (49 %) stage 1 hypertension, and 6 (5 %) stage 2 hypertension. With the observed prevalence of WCH, initial ambulatory BP monitoring use yielded net savings after evaluation of 3 patients, with projected savings of \$2.4 million per 1,000 patients. The authors concluded that ambulatory BP monitoring in the initial evaluation of suspected childhood hypertension is highly cost-effective. Awareness of cost saving potential may increase the availability of ambulatory BP monitoring for evaluation of new-onset hypertension.

Muxfeldt et al (2012) stated that resistant hypertension is defined as uncontrolled office BP, despite the use of greater than or equal to 3 anti-hypertensive drugs. Ambulatory BP monitoring (ABPM) is mandatory to diagnose 2 different groups, those with true and white-coat resistant hypertension. Patients are found to change categories between controlled/uncontrolled ambulatory pressures without changing their office BP.

In this way, ABPM should be periodically repeated. The aim of this study was to evaluate the most appropriate time interval to repeat ABPM to assure sustained BP control in patients with white-coat resistant hypertension. This prospective study enrolled 198 patients (69 % women; mean age of 68.9 +/- 9.9 years) diagnosed as white-coat resistant hypertension on ABPM. Patients were submitted to a second confirmatory examination 3 months later and repeated twice at 6-month intervals. Statistical analyses included Bland-Altman repeatability coefficients and multi-variate logistic regression. Mean office BP was 163 ± 20/84 ± 17 mm Hg, and mean 24-hour BP was 118 ± 8/66 ± 7 mm Hg. White-coat resistant hypertension diagnosis presented a moderate reproducibility and was confirmed in 144 patients after 3 months. In the 3rd and 4th ABPMs, 74 % and 79 % of patients sustained the diagnosis. In multi-variate regression, a daytime systolic blood pressure less than or equal to 115 mm Hg in the confirmatory ABPM triplicated the chance of white-coat resistant hypertension status persistence after 1 year. The authors concluded that a confirmatory ABPM is necessary after 3 months of the 1st white-coat-resistant hypertension diagnosis, and the procedure should be repeated at 6-month intervals, except in patients with daytime systolic blood pressure less than or equal to 115 mm Hg, in whom it may be repeated annually.

Vollebregt and colleagues (2013) stated that it is not known whether automated devices for measuring BP perform better than conventional sphygmomanometry in predicting preeclampsia. In a prospective, observational, cohort study, these investigators compared 2 different automated devices with conventional sphygmomanometry for their association with development of preeclampsia or gestational hypertension. A total of 289 healthy normotensive women of whom 235 were nulliparous and 44 parous with preeclampsia in a previous pregnancy were included in this study. At 8 to 11 weeks of pregnancy, BP was measured with 2 different automated devices (continuous finger arterial pressure waveform registration and ABPM) and with conventional sphygmomanometry. Main outcome measures were preeclampsia and gestational hypertension. Blood pressure in the 1st trimester, as measured with all 3 methods, was significantly higher in women who developed preeclampsia or gestational hypertension. After adjustment for previous preeclampsia, the point estimate of the odds ratios for association with later preeclampsia for both automated devices were comparable and higher than for conventional sphygmomanometry; however, differences were not statistically significant. The odds ratio (95 % confidence intervals) for every 1 mmHg pressure increase of mean arterial pressure was 1.08 (1.02 to 1.15) for sphygmomanometry, 1.17 (1.09 to 1.27) for finger arterial pressure waveform registration, and 1.17 (1.07 to 1.27) for ABPM. Results were comparable if preeclampsia and gestational hypertension were analyzed together. The authors concluded that BP in the 1st trimester was associated with the development of hypertensive disorders of pregnancy; however, no significant differences were found between measurements by automatic devices including ABPM compared with conventional sphygmomanometry.

O'Brien and Dolan (2016) reviewed the current recommendations for ABPM and the use of ABPM in assessing treatment. These investigators reviewed current international guidelines and undertook a critical review of evidence supporting the clinical use of ABPM in effectively managing anti-hypertensive drug treatment. Current guidelines emphasize the diagnostic superiority of ABPM, mainly from the ability of the technique to identify sustained hypertension by allowing for the exclusion of white-coat hypertension and by demonstrating the presence of masked hypertension; ABPM also offers diagnostic

insights into nocturnal patterns of BP, such as dipping and non-dipping, reverse dipping, and excessive dipping, and the presence of nocturnal hypertension; although less attention is given to the nocturnal behavior of BP in clinical practice, the nocturnal patterns of BP have particular relevance in assessing the response to BP-lowering medication. Surprisingly, although the current guidelines give detailed recommendations on the diagnostic potential and use of ABPM, there are scant recommendations on the benefits and application of the technique for the initiation of BP-lowering therapy in clinical practice and virtually no recommendations on how it might be used to assess the effectiveness of drug treatment. The authors concluded that in view of a deficiency in the literature on the role of ABPM in assess the effectiveness of drug treatment, they put forward proposals to correct this deficiency and guide the prescribing physician on the most appropriate drug administration and dosage over time.

Routine Use Before Initiation of Continuous Positive Airway Pressure (CPAP) Therapy

Castro-Grattoni and colleagues (2017) stated that the reduction in BP with continuous positive airway pressure (CPAP) is modest and highly variable. These researchers identified the variables that predict BP response to CPAP; 24-hour ABPM, C-reactive protein (CRP), leptin, adiponectin and 24-hour urinary catecholamine were measured before and after 6 months of CPAP in obstructive sleep apnea (OSA) patients. A total of 88 middle-aged, obese male patients with severe OSA (median apnea-hypopnea index [AHI] of 42 events/hour) were included; 28.4 % had hypertension; 62 patients finished the study, and 60 were analyzed. The daytime DBP (-2 mmHg) and norepinephrine (-109.5 nmol/day) were reduced after CPAP, but no changes in the 24-hour BP, night-time BP, dopamine, epinephrine, CRP, leptin or adiponectin were detected. The nocturnal normotension was associated with an increased night-time-BP (+4 mmHg) after CPAP, whereas nocturnal hypertension was associated with a reduction of 24-hour BP (-3 mmHg). A multi-variate linear regression model showed differential night-time BP changes after CPAP. Specifically, low night-time heart rate (HR; less than 68 bpm) and BP dipper profile were associated with increased night-time BP and new diagnosis of nocturnal hypertension. The authors concluded that these findings suggested that nocturnal hypertension, circadian BP pattern and night-time HR could be clinical predictors of BP response to CPAP and support the usefulness of 24-hour ABPM for OSA patients before treatment initiation. Moreover, they stated that these findings need to be confirmed in further studies.

Accuracy of Automated BP Measurements in Patients with Atrial Fibrillation

Clark and colleagues (2019) noted that atrial fibrillation (AF) affects approximately 3 % of the general population and is twice as common with hypertension. Validation protocols for automated sphygmomanometers exclude people with AF, raising concerns over accuracy of hypertension diagnosis or management, using out-of-office BP monitoring, in the presence of AF. Some devices include algorithms to detect AF; a feature open to misinterpretation as offering accurate BP measurement with AF. These researchers examined the accuracy of automated devices, with or without AF detection, for measuring BP. They searched Medline and Embase to October 2018 for studies comparing automated BP measurement devices to a standard mercury sphygmomanometer contemporaneously. Data were extracted by 2 reviewers. Mean BP differences between devices and mercury were calculated, where not reported and compared; meta-analyses were undertaken where possible. These investigators included 13 studies reporting 14 devices. Mean systolic and diastolic BP differences from mercury ranged from -3.1 to + 6.1/-4.6 to +9.0 mmHg. Considerable heterogeneity existed between devices (I²: 80 to 90 %). Devices with AF detection algorithms appeared no more accurate for BP measurement with AF than other devices. A previous review concluded that oscillometric devices were accurate for systolic but not diastolic BP measurement in AF. The findings of this study did not support that conclusion. As a consequence of heterogeneity between devices, they should be evaluated on individual performance. The authors found no evidence that devices with AF detection measured BP more accurately in AF than other devices. These researchers stated that more home or ambulatory automated BP monitors require validation in populations with AF.

Comparison Between Automated Office Blood Pressure Measurements and Manual Office Blood Pressure Measurements

Bo et al (2021) stated that automated office BP (AOBP) measurements may provide more accurate estimation of BP than manual office BP (MOBP) measurements. In a systematic review, these investigators examined the diagnostic performance of AOBP and MOBP using ABPM as reference. Several databases including Medline, Embase, Scopus, and China

Academic Journals were searched. Data were extracted, double-checked by 2 investigators, and were analyzed using a random effects model. A total of 26 observational studies were included. The mean systolic/diastolic BP obtained by AOBP was not significantly different from that obtained by ABPM. The sensitivity and specificity of AOBP to detect elevated BP were approximately 70 %. Fewer participants had white-coat hypertension on AOBP measurement than on MOBP measurement (7 % versus 14 %); however, about 13 % had masked hypertension on AOBP measurement. The width of the limit of agreement comparing AOBP and ABPM; and MOBP and ABPM was comparable. The authors concluded that AOBP may reduce the rate of the observed white-coat effect but undermined masked hypertension. The current recommendation, however, is limited by the absence of high-quality studies and the high heterogeneity of these findings. These researchers stated that more high-quality studies using different AOBP machines and in different population are needed.

Viera et al (2021) noted that office BP measurements are not the most accurate method to diagnose hypertension. Home BP monitoring (HBPM) and 24-hour ABPM are out-of-office alternatives, and ABPM is considered the reference standard for BP assessment. In a systematic review, these investigators examined the accuracy of oscillometric office and home BP measurement methods for correctly classifying adults as having hypertension, defined using ABPM. Data sources included PubMed, Cochrane Library, Embase, ClinicalTrials.gov, and DARE databases and the American Heart Association (AHA) website (from inception to April 2021) were searched, along with reference lists from retrieved articles. Two authors independently abstracted raw data and evaluated methodological quality; a 3rd author resolved disputes as needed. Random effects summary sensitivity, specificity, and likelihood ratios (LRs) were calculated for BP measurement methods for the diagnosis of hypertension. ABPM (24-hour mean BP of greater than or equal to 130/80 mm Hg or mean BP while awake of greater than or equal to 135/85 mm Hg) was considered the reference standard. A total of 12 cross-sectional studies (n = 6,877) that compared conventional oscillometric office BP measurements to mean BP during 24-hour ABPM and 6 studies (n = 2,049) that compared mean BP on HBPM to mean BP during 24-hour ABPM were included (range of 117 to 2,209 participants per analysis); 2 of these studies (n = 3,040) used consecutive samples. The overall prevalence of hypertension identified by 24-hour ABPM was 49 % (95 % CI: 39 % to 60 %) in the pooled studies that evaluated office measures and 54 % (95 % CI: 39 % to 69 %) in studies that evaluated HBPM. All included studies assessed sensitivity and specificity at the office BP threshold of 140/90 mm Hg and the home BP threshold of 135/85 mm Hg. Conventional office oscillometric measurement (1 to 5 measurements in a single visit with BP of greater than or equal to 140/90 mm Hg) had a sensitivity of 51 % (95 % CI: 36 % to 67 %), specificity of 88 % (95 % CI: 80 % to 96 %), positive LR of 4.2 (95 % CI: 2.5 to 6.0), and negative LR of 0.56 (95 % CI: 0.42 to 0.69). Mean BP with HBPM (with BP of greater than or equal to 135/85 mm Hg) had a sensitivity of 75 % (95 % CI: 65 % to 86 %), specificity of 76 % (95 % CI: 65 % to 86 %), positive LR of 3.1 (95 % CI: 2.2 to 4.0), and negative LR of 0.33 (95 % CI: 0.20 to 0.47). Two studies (1 with a consecutive sample) that compared unattended automated mean office BP (with BP of greater than or equal to 135/85 mm Hg) with 24-hour ABPM had sensitivity ranging from 48 % to 51 % and specificity ranging from 80 % to 91 %. One study that compared attended automated mean office BP (with BP of greater than or equal to 140/90 mm Hg) with 24-hour ABPM had a sensitivity of 87.6 % (95 % CI: 83 % to 92%) and specificity of 24.1 % (95 % CI: 16 % to 32 %). The authors concluded that office measurements of BP may not be accurate enough to rule in or rule out hypertension; HBPM may be helpful to confirm a diagnosis. These investigators stated that when there is uncertainty regarding threshold values or when office and HBPM are not in agreement, 24-hour ABPM should be considered to establish the diagnosis.

Smartphone Application-Assisted Home Blood Pressure Monitoring Compared with Office and Ambulatory Blood Pressure Monitoring in Patients with Hypertension

Groenland et al (2022) noted that the development of automated, smartphone application (app)-assisted HBPM allows for standardized measurement of BP at home. In an open, randomized, 5-way cross-over study, these researchers examined the (diagnostic) agreement between app-assisted HBPM, automated office BP (OBP), and the reference standard 24-hour ABPM. Patients diagnosed with hypertension were randomized to 1 of 10 clusters, each containing 5 BP measurement methods (ABPM, HBPM, attended OBP, unattended OBP, and unattended 30-min BP) in different order. A total of 113 patients were included. The average 24-hour ABPM was 126 ± 11 / 73 ± 8 mm Hg compared with 141 ± 14 / 82 ± 10 mm Hg with app-assisted HBPM, 134 ± 13 / 80 ± 9 mm Hg with unattended 30-min BP, 137 ± 16 / 81 ± 11 mm Hg with attended OBP, and 135 ± 15 / 81 ± 10 mm Hg with unattended OBP monitoring. Diagnostic agreement between app-assisted HBPM and 24-hour ABPM for

diagnosing sustained (OBP greater than 140/90 mm Hg and ABPM of greater than or equal to 130/80 mm Hg or HBPM of greater than or equal to 135/85 mm Hg), white-coat (OBP of greater than or equal to 140/90 mm Hg and ABPM of less than 130/80 mm Hg or HBPM of less than 135/85 mm Hg), and masked hypertension (OBP of less than 140/90 mm Hg and ABPM of greater than or equal to 130/80 mm Hg or HBPM of greater than or equal to 135/85 mm Hg) was fair-to-moderate (κ statistics ranging from 0.34 to 0.40). App-assisted HBPM had high sensitivities (78 % to 91 %) and negative predictive values (90 % to 97 %) for diagnosing sustained and masked hypertension. The authors concluded that this study showed a considerable (diagnostic) disagreement between app-assisted HBPM and ABPM. They noted that the high sensitivity and negative predictive value (NPV) for diagnosing sustained and masked hypertension suggested that app-assisted HBPM may be suitable for screening on these hypertension phenotypes, but not a replacement of, ABPM.

The authors stated that this study had 2 main drawbacks. First, since the majority of the population was prescribed anti-hypertensive medication, the findings may not be generalizable to individuals (in primary care) who have not yet been prescribed anti-hypertensive medications, in whom BP variability was likely to be greater. Second, the cross-over design of this study did not allow the performance of HBPM to be evaluated based on the occurrence of clinical outcomes, which is considered an important aspect in determining the best BP measurement method.

Autosomal Dominant Polycystic Kidney Disease

Patil et al (2015) noted that the use of ABPM to evaluate BP in patients with autosomal dominant polycystic kidney disease (ADPKD) has resulted in greater precision and showed abnormalities of BP in ADPKD patients. Almost 33 % of children with ADPKD showed exclusively nocturnal hypertension. A significant proportion of normotensive young adults with ADPKD have “pre-hypertension” by ABPM. Blunted “nocturnal dipping” on ABPM has also been reported to be associated with endothelial dysfunction in this population.

Dudley et al (2019) stated that ADPKD causes a progressive decline in renal function, with kidney failure tending to occur in middle age. Children and young individuals with ADPKD may not have any symptoms; however they may have high BP, which may accelerate progression to later stages of chronic kidney disease. These investigators noted that there is uncertainty and variation in how health professionals manage children and young people with confirmed or a family history of ADPKD, because of a lack of evidence. For example, health professionals may be unsure regarding when to test children’s BP and how often to monitor it in the hospital clinic or at the general practitioner (GP)’s office. These researchers developed guideline to help families affected by ADPKD by making sure that BP assessment is undertaken regularly in children and young individuals at risk of developing ADPKD.

On behalf of the Network for Early Onset Cystic Kidney Disease (NEOCYST), Gimpel et al (2019) developed recommendations on the diagnosis and management of ADPKD in children and young individuals. They were endorsed by the International Pediatric Nephrology Association (IPNA) and the European Society of Paediatric Nephrology (ESPN). For asymptomatic minors at risk of ADPKD, ongoing surveillance (repeated screening for treatable disease manifestations without diagnostic testing) or immediate diagnostic screening are equally valid clinical approaches. Ultrasonography (US) is the current radiological method of choice for screening. Sonographic detection of 1 or more cysts in an at-risk child is highly suggestive of ADPKD; however, a negative scan could not rule out ADPKD in childhood. Genetic testing is recommended for infants with very-early-onset symptomatic disease and for children with a negative family history and progressive disease. Children with a positive family history and either confirmed or unknown disease status should be monitored for hypertension (preferably by ABPM) and albuminuria. Currently, vasopressin antagonists should not be offered routinely; however, off-label use could be considered in selected children. No consensus was reached on the use of statins; however, mammalian target of rapamycin (mTOR) inhibitors and somatostatin analogs are not recommended. Children with ADPKD should be strongly encouraged to achieve the low dietary salt intake that is recommended for all children.

Benz and Hartung (2021) stated that ADPKD and autosomal recessive polycystic kidney disease (ARPKD) are characterized by bilateral cystic kidney disease resulting in progressive decline in renal function. These diseases also have distinct liver manifestations. The range of clinical presentation and severity of both ADPKD and ARPKD is much wider than was once recognized. Pediatric and adult nephrologists are likely to care for individuals with both diseases in their lifetimes. These investigators noted

that hypertension is common in patients with ADPKD, and often precedes a decline in glomerular filtration rate (GFR). Approximately 35 % of children and 80 % of adults with ADPKD exhibit high BP, with a mean age of diagnosis of hypertension of around 39 years for patients with PKD1 mutations, and around 49 years for those with PKD2 mutations. Nocturnal hypertension is also very common in children with ADPKD: in an ABPM study of 310 children (mean age of 11.5 years), 52 % of the cohort lacked the physiologic nocturnal dip, and 18 % had isolated nocturnal hypertension. Hypertension is associated with higher kidney volumes and faster renal function decline in children and adults with ADPKD.

GeneReviews' webpage on "Polycystic kidney disease, autosomal dominant" (Harris and Torres, 2022) stated that "Hypertension is often diagnosed much later than when it first occurs in individuals with ADPKD; 24-hour monitoring of ambulatory blood pressure of children or young adults may reveal elevated blood pressure, attenuated decrease in nocturnal blood pressure, and exaggerated blood pressure response during exercise, which may be accompanied by left ventricular hypertrophy and diastolic dysfunction. Monitoring of blood pressure in children at risk for ADPKD is recommended".

Cadnapaphornchai and Ong (2023) stated that ADPKD is the most common hereditary kidney disease and a major cause of renal failure globally. Significant kidney enlargement occurs decades preceding loss of renal function. However, the earliest clinical manifestations of disease have been less well characterized in young adults, a typically healthy population who do not often seek routine medical care. These investigators stated that in children with ADPKD, 24-hour ABPM has been shown to be a valuable tool. Using ABPM, 35 % of children with ADPKD were shown to have a lack of BP dipping, and 18 % showed isolated nocturnal hypertension, an important predictor of target organ damage. Although fewer studies have been conducted in young adults (18 to 30 years) with ADPKD, ABPM has value for detection of masked hypertension in this group. Young adults with ADPKD should be followed closely from diagnosis, with routine BP monitoring ideally using ABPM, with careful assessment of associated extra-renal disease.

Furthermore, an UpToDate review on "Autosomal dominant polycystic kidney disease (ADPKD): Evaluation and management of hypertension" (Chapman et al, 2024) states that "Use of 24-hour ambulatory blood pressure monitoring (ABPM) -- Ambulatory blood pressure may be helpful to diagnose hypertension and is indicated in untreated (and especially young adults) with borderline high office blood pressure and suspicion of white-coat hypertension".

Appendix

Blood Pressure Levels by Age and Height Percentile: [Blood Pressure Levels for Boys and Girls by Age and Height Percentile](#) 

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The above policy is based on the following references:

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